

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. True/False: An ergodic process will always have the same stochastic characteristics for every location tof the random process X(t).

1 / 1 point

- ☐ True
☒ False

✔ Correct
Correct! An ergodic process has the same stochastic characteristics from one sample to the next, but there is no guarantee that an ergodic process has the same stochastic characteristics across t for any one given sample.

2. True/False: A stationary process will always have the same stochastic characteristics for every location t of the random process X(t).

1 / 1 point

- ☒ True
☐ False

✔ Correct
By definition, a stationary (or statistically homogeneous) process has the same stochastic characteristics across t for any one given sample.

3. True/False The marginal PDF of a random partition will always be continuous.

1 / 1 point

- ☐ True
☒ False

✔ Correct
Correct! A random partition is a random process that takes on discrete values. Discrete random variables are described by probability mass functions, which are not continuous PDFs.

4. Given a CDF $F_X(x) = 1 - e^{-x}; x > 0$ what is the probability that $X < 2$?

1 / 1 point

- ☐ 0
☐ 1
☐ 0.14
☒ 0.86

✔ Correct
Correct! The probability that X<2 is equal to the CDF evaluated at x=2, or F(2)=1-e^(-2)=0.86.

5. Given a PDF $f_X(x) = 3x^2; 0 < x < 1$ what is the probability that $X < 0.5$?

1 / 1 point

- ☐ 0.75
☐ 0.5
☒ 0.125
☐ 0.25

✔ Correct
Correct! The probability that X<0.5 is equal to the CDF evaluated at x=0.5. The CDF is the integral of the PDF from 0 to x, or F(x)=x^3. Therefore P(X<0.5)=F(0.5)=0.5^3=0.125.

6. Given the same PDF $f_X(x) = 3x^2; 0 < x < 1$ what is the expected value (mean) of X ?

1 / 1 point

- ☒ 0.75
☐ B: 0.5
☐ 0.125
☐ D: 0.25

✔ Correct
Correct! The expected value is the integral of $x * f_X(x)$ over the domain of $0 < x < 1$. The integral of $3x^3$ is $3/4x^4$. Substituting in $x = 1$ and $x = 0$, the mean value is therefore ¾ or 0.75.

7. Given the same PDF $f_X(x) = 3x^2; 0 < x < 1$ what is the variance of X ?

1 / 1 point

- ☐ 1.0
☒ *B: 0.0375
☐ 0.5625
☐ D: 0.25

✔ Correct
Correct! The variance is the expected value of X^2 , minus the square of the expected value of X (which we know from the previous question is 0.75). The expected value of X^2 is the integral of $3x^4$ from 0 to 1, or $3/5$. Therefore the variance is $3/5 - 0.75^2 = 0.0375$.

8. Consider a Gaussian random process (i.e., a random process with a Gaussian marginal distribution), with mean 1 and standard deviation 2. If you sample the process at a large number of finely spaced points, what fraction of the sampled values would you expect to be less than 3?

1 / 1 point

- ☒ About 84%
☐ About 16%
☐ About 98%
☐ 0%

✔ Correct
Correct! We can find the CDF of a Gaussian random variable by first normalizing the number by its mean and standard deviation. In this case, the normalized Gaussian random variable is (3-1)/2=1. A table of standard normal Gaussian random variables will indicate that the corresponding probability is about 0.84.